ULTRASOUND ON A CHIP Supercharged With AI FOR MEDICAL SCANNING

About two-thirds of the world's population lacks access to medical imaging, whether in developing nations or communities living in underdeveloped nations. Built on a chip, Butterfly Network has been providing solution to the above said problem. This is through its specifically designed 3D ultrasound imaging system on a chip (SoC)

The author, **VINAYAK RAMACHANDRA ADKOLI**, is BE in industrial production and has served as a lecturer in three different polytechnics for ten years. He is also a freelance writer and cartoonist

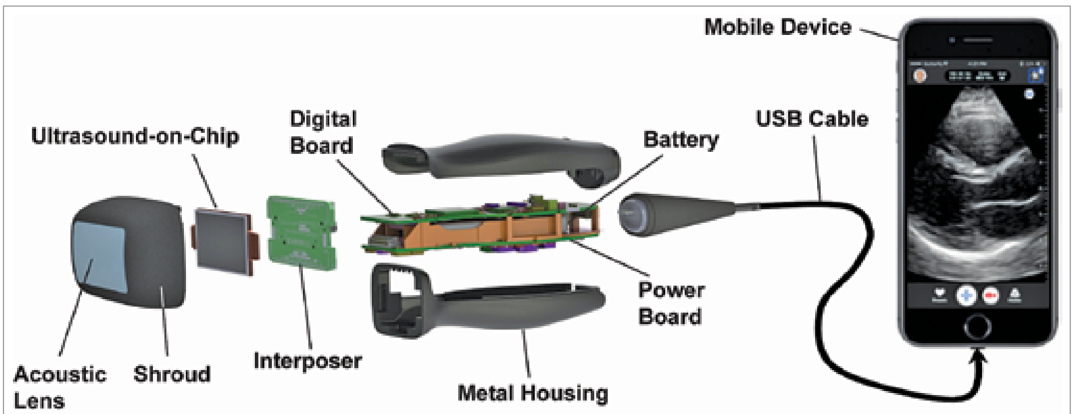
Diagnostic ultrasound, also called sonography or diagnostic medical sonography, is an imaging method that uses sound waves to produce images of organs within your body. The images can provide valuable information for diagnosing and directing treatment for a variety of diseases and conditions

Most ultrasound scannings are done outside the body but some are done by placing a small probe inside the body. Traditional ultrasound scanning devices used by radiologists work via piezoelectric crystals. Here an oscillating crystal produces sound waves that generate an image. At present, such scanning probes cost $8,000 while the traditional ultrasound unit itself costs $100,000 or more.

Nowadays there is much discussion about Butterfly Network and its iQ ultrasound probe. This probe is a nano, handy, and point-of-care ultrasound probe in a saturated ultrasound scanning market. The Butterfly Network has found a way to generate these sound waves with semiconductor chips.

There are massive efficiencies associated with the semiconductor fabrication process, which have already made the iQ probe the cheapest device on the market and costs just $1,999. More importantly, however, the semiconductor approach means the iQ probe will follow Moore's Law and will become massively cheap and better with time. This semiconductor approach also gives the iQ probe a massive acoustic bandwidth, a critical detail that makes it uniquely portable.

The narrow bandwidth of traditional ultrasound means that a variety of probes are needed. For example, one probe for the deep structures of the abdomen and another for the superficial thyroid gland. You thus have to transport a large device and/or multiple probes around the hospital.



Working of Ultrasound-on-Chip in a portable ultrasonic scanner (Credit: Google Images; https://www.pnas.org)